Alcohol-Related Seizures

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persistent heavy drinking ⇒ seizures independent of withdrawal.

KEYWORDS

• Seizures • Alcohol withdrawal • Benzodiazepines

Among the various medical problems related to alcohol abuse, the differential diagnosis and management of seizures remains the most challenging and controversial problem. Any patient arriving to the emergency department (ED) with seizures should be questioned about alcohol intake. From 20% to 40% of patients with seizures who present to an ED have seizures related to alcohol abuse.[1] Alcohol is a causative factor in 15% to 24% of patients with status epilepticus.[2] The primary consideration in the initial care of these patients is the recognition and treatment of life-threatening causes of seizures, such as central nervous system (CNS) infection, hypoglycemia, and intracranial hematoma.

Alcohol may act in one of the several ways to produce seizures in patients with or without underlying foci:

- By partial or absolute withdrawal of alcohol after a period of chronic intake.
- By an acute alcohol-related metabolic brain disorder (eg, hypoglycemia, hyponatremia).
- By creating a situation leading to cerebral trauma.
- By precipitating seizures in patients with idiopathic or posttraumatic epilepsy.
- Persistent heavy intake of alcohol, independent of alcohol withdrawal.

Moreover, alcoholic patients are more susceptible to other disorders associated with seizures, including cerebral infarction, neurosyphilis, AIDS, brain abscess, meningitis, cerebral trauma, and subarachnoid hemorrhage (**Box 1**).[3–5]

ALCOHOL WITHDRAWAL SEIZURES

Chronic alcohol consumption affects central α-adrenergic and β-adrenergic receptors, the inhibitory neurotransmitter γ-aminobutyric acid, and dopamine turnover. The hallmark of alcohol withdrawal is CNS excitation with increased levels of cerebrospinal fluid (CSF) and plasma and urinary catecholamine.[6,7] The withdrawal syndrome develops 6 to 24 hours after the reduction of ethanol intake and usually

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Box 1
Differential diagnosis of alcohol-related seizures

Withdrawal (alcohol or drugs)

Exacerbation of an underlying seizure disorder

Acute drug intoxication (eg, amphetamines, anticholinergics, cocaine, isoniazid, organophosphates, tricyclics, salicylates)

Metabolic disorders (eg, hypoglycemia, hyponatremia, hypernatremia, hypocalcemia)

Infections (CNS or systemic)

Trauma (acute or remote)

Stroke

Noncompliance with anticonvulsants

lasts for 2 to 7 days. The alcohol withdrawal state ranges from mild withdrawal with insomnia and irritability to major withdrawal with diaphoresis, fever, disorientation, and hallucinations.[8]

Minor alcohol withdrawal syndrome occurs as early as 6 hours and usually peaks at 24 to 36 hours after the cessation or significant decrease in alcohol intake. It is characterized by mild autonomic hyperactivity including nausea, anorexia, tremor, tachycardia, hypertension, hyperreflexia, sleep disturbances (eg, insomnia, vivid dreams), and anxiety.[9] Major alcohol withdrawal syndrome occurs after more than 24 hours and usually peaks at 50 hours but occasionally takes up to 5 days after the decline or termination of drinking. This syndrome is characterized by anxiety that is more pronounced, insomnia, irritability, tremor, anorexia, tachycardia, hyperreflexia, hypertension, fever, decreased seizure threshold, auditory and visual hallucinations, and finally, delirium.[8–10]

Delirium tremens is at the extreme end of the alcohol withdrawal spectrum and is characterized by gross tremor, profound confusion, fever, incontinence, frightening visual hallucinations, and mydriasis. It seldom occurs before the third postabstinence day. Only 5% of patients hospitalized for alcohol withdrawal develop delirium tremens. True delirium tremens is rare and is not synonymous with alcohol withdrawal. Alcohol withdrawal seizures (AWS) can occur in minor alcohol withdrawal, major alcohol withdrawal, or delirium tremens.[9,10]

Alcohol withdrawal is one of the most common causes of adult-onset seizures. Descriptions of AWS were based on data collected by Victor and Brausch[11] on 241 alcohol abusers with seizures or alcohol-related illness complicated by seizures. These patients with AWS were confirmed to be alcoholic for many years, with the onset of seizures in adulthood. The seizures occurred 6 to 48 hours after the cessation of drinking. About 90% of the patients had 1 to 6 generalized tonic-clonic seizures, which occurred without warning, whereas 60% experienced multiple seizures, usually 2 to 4 within a 6-hour period. However, recent data suggest a much lower repeat seizure rate of 13% to 24%.[12] The incidence of partial (focal) seizures, common with posttraumatic epilepsy, is increased by alcohol withdrawal. However, partial seizures indicate a mass lesion until proven otherwise.

The term AWS is reserved for seizures with the characteristics described by Victor and Brausch.[11] The term alcohol-related seizures (ARS) is used to refer to all seizures in the aggregate associated with alcohol use, including the subset of AWS.

PATIENTS PRESENTING WITH A NORMAL NEUROLOGIC EXAMINATION

New-Onset ARS

Patients with new-onset ARS must be thoroughly evaluated, including alcoholic patients who claim to have had seizures in the past but for whom no documentation of previous seizures or of an appropriate workup is available. Possibility of metabolic disorders, toxin ingestion, infection, and structural abnormalities is ruled out by history, physical examination, and laboratory testing, including the estimation of levels of electrolytes, serum urea nitrogen, creatinine, glucose and computed tomographic (CT) scan, as necessary. An electroencephalography scheduled during follow-up may be considered (but its value is limited in the patient with unexplained seizures.[13]

If the results of initial physical examination and laboratory tests are normal, patients who remain seizure free and symptom free with no sign of withdrawal after 4 to 6 hours of observation may be discharged. Optimal outpatient treatment include clear guidelines for follow-up and reevaluation and the help of a concerned family member or friend (who is not a drinking partner) who should remain with the patient for 1 to 2 days. These criteria may be difficult to meet, and the physician must use discretion in deciding to admit for observation when the patient is at risk of serious injury. The ideal disposition is participation in a detoxification/rehabilitation program.

It may be unclear whether the patient has had AWS or the new onset of a seizure disorder in the setting of alcohol consumption. At this point, such a patient probably does not require further treatment. The literature provides little information about the natural history, including the risk of subsequent seizures, in patients in the ED, presenting with an unprovoked seizure. Beginning anticonvulsant treatment in a patient after a new-onset single seizure is controversial, and the final decision regarding treatment should be individualized and made after consultation with a neurologist or the patient's primary care provider. The patient with first-time ARS, who has a history that is consistent with AWS and a workup with negative results can be treated as presented in the following sections.[14]

Alert patients with a history of prior seizures, during alcohol withdrawal

Alcoholic patients with the manifestations of alcohol withdrawal, who have not had a recent seizure but have a related history of AWS are encountered frequently in the ED. The risk of seizure increases 10-fold in this subset of patients.[15,16]

Benzodiazepines alone are sufficient to prevent AWS.[12] Many of patients with AWS sporadically take one or more anticonvulsants. It is difficult, if not impossible, to determine in the ED whether these anticonvulsants were given for an underlying seizure disorder or AWS. To effectively prevent AWS, detoxification with benzodiazepines should be initiated early, because most AWS occur within the first 24 hours after alcohol withdrawal. Treatment should be started with the understanding that the patient will be observed for 4 to 6 hours and referred to a detoxification/rehabilitation program (if available).

An initial test dose of 2 mg of lorazepam or 10 mg of diazepam can be given orally or 1 mg of lorazepam or 5 mg of diazepam intravenously to the patient in the ED. This dose may be repeated depending on the patient's response. The patient is observed for 4 to 6 hours, which guides the dose required for outpatient treatment. Prescribing anticonvulsants, such as benzodiazepines (other than a short 3- to 6-day course for alcohol withdrawal) and phenytoin, to the discharged alcoholic patient may increase the potential for addiction or may paradoxically increase the number of acute seizures. For a concurrent seizure disorder, the poorly compliant alcoholic patient may do better without anticonvulsants prescribed for outpatients.[15] Therefore, the ideal disposition is admission to a detoxification/rehabilitation unit.

Alert patients with a seizure, before or after presentation

Alcoholic patients with a documented history of ARS, who experience a single seizure or a short burst of seizures should be treated with lorazepam, 2 mg intravenously. These patients usually require observation with careful monitoring of neurologic status for at least 6 hours, before discharge.[12]

PATIENTS WITH AN ABNORMAL NEUROLOGIC PRESENTATION

New-Onset Partial Seizure

Partial (focal) seizures are reported to account for up to 24% of ARS. Conversely, studies have shown that 17% to 21% of patients with partial ARS have structural lesions (hematomas, tumors, or vascular abnormalities).[17] These primary causes of partial ARS, such as prior head trauma, may be missed in the history taking. As a result, an emergent CT scan is indicated to evaluate new-onset partial seizures.

The patient with a documented history of focal ARS, who has been previously evaluated does not require an emergency CT scan, provided a return to baseline occurs promptly. A patient presenting with focal ARS with subsequent normal neurologic imaging can be managed with supportive care, observation for 6 hours, and treatment with benzodiazepine for withdrawal signs or symptoms. Appropriate follow-up should be arranged.

MANAGEMENT

Historically, up to one-third of patients with AWS progressed to delirium tremens because of inadequate treatment. Currently, this proportion has decreased to less than 5% with early aggressive benzodiazepine therapy.[18]

An intravenous line of normal saline should be established. If the patient has an altered mental status, administration of thiamine, magnesium, dextrose, and naloxone should be considered. Empirical glucose bolus dosing should not be prescribed if a prompt and accurate determination of blood glucose levels is possible.[19]

Although magnesium does not decrease the severity of withdrawal symptoms or the incidence of delirium or seizures, it carries no significant risk or cost. In the nonacute setting, oral magnesium supplementation in long-term alcoholic patients improves liver function test results, electrolyte balance, and muscle strength.[20] Intravenous administration of multivitamin preparations may be considered for chronic malnutrition but their clinical benefit is not proven.

Obtundation

The obtunded or stuporous alcohol-dependent patient with a history of seizure poses a diagnostic challenge. The patient's decreased level of consciousness (LOC) may be because of a postictal state, an occult head trauma, an unrecognized metabolic disorder, or poisoning. The first task is to determine the possibility of hypoglycemia (diagnosed and reversed at the bedside in seconds to minutes) or a neurosurgical lesion with morbidity and mortality directly related to the time of surgical intervention.

Patients with focal neurologic findings on physical examination, new-onset seizures, partial (focal) seizures, or evidence of acute head trauma should be considered for an urgent CT scan. If the LOC is consistently improving, the patient is unlikely to have an immediate life-threatening problem. An unimproved or deteriorating LOC requires an urgent CT scan.

No History of Seizure/No Current Seizure

In the patient with alcohol withdrawal who lacks a history of seizures, benzodiazepines generally have sufficient anticonvulsant activity to prevent withdrawal seizures.[21]

Status Epilepticus

Although fewer than 8% of patients with ARS progress to status epilepticus, alcohol is implicated in 15% to 24% of patients with status epilepticus.[22] Status epilepticus may also be the first presentation of ARS. The most common cause of status epilepticus is discontinuation of or erratic compliance with an anticonvulsant drug regimen, followed by ARS. However, status epilepticus may occur for a variety of reasons and is often multifactorial. Thus, it is essential to screen for all possible factors underlying repeated or prolonged seizures, even when a probable cause is thought to be readily apparent.

Initial interventions for the alcoholic in status epilepticus include stabilization of the airway, breathing, and circulation; administration of glucose and thiamine as indicated; and treatment with a benzodiazepine. Lorazepam and diazepam are both effective in terminating seizures in status epilepticus. Lorazepam is preferable because its anticonvulsant effect lasts several hours, whereas diazepam's anticonvulsant effect lasts only from 20 to 30 minutes. Because of this short-term effect lorazepam has fewer recurrent seizures and requires fewer repeat doses than diazepam.[23–25]

If seizures persist after lorazepam or diazepam administration, a loading dose of phenytoin, 20 mg/kg, or fosphenytoin, 15 to 20 phenytoin equivalent mg/kg, is administered in the second intravenous line. Additional doses of 5 mg/kg of phenytoin, to a maximum of 30 mg/kg, can be given if status is not terminated.[23]

Most of the cardiac abnormalities associated with phenytoin infusion are caused by its solvent propylene glycol. Caution is advised in patients who have preexisting heart disease. Using a 20-gauge intravenous line proximal to the forearm to avoid purple glove syndrome and keeping the phenytoin rate less than 50 mg/min can minimize the problem of hypotension and bradycardia. Fosphenytoin does not contain propylene glycol and is safe and well tolerated at intravenous rates of 100 to 150 mg/min.

Rapid achievement of therapeutic free phenytoin levels, without significant side effects, has been documented using fosphenytoin in patients with status epilepticus.[26] However, the benefit of fosphenytoin versus phenytoin in the ED has been questioned.[26–30]

Phenobarbital was recommended in the past for patients who fail to respond to benzodiazepine and phenytoin. Experts have suggested skipping phenobarbital for refractory status and initiating infusion of pentobarbital, propofol, or midazolam.[23,31–35]

It is often difficult to rule out a CNS infection in alcoholic patients with status epilepticus because of concomitant hyperthermia, serum leukocytosis, and CSF pleocytosis. If CSF infection is a possibility, intravenous antibiotic therapy should be started, blood cultures obtained, and a lumbar puncture considered.[36]

PHENYTOIN/ANTICONVULSANT CONUNDRUM

Phenytoin has no significant benefit over placebo in preventing recurrence of AWS.[1,37] Considering the risks of phenytoin and no demonstrated benefit in the setting of AWS, it is not indicated for the treatment of AWS. The sudden withdrawal of phenytoin administration may induce the convulsive effects of alcohol withdrawal. Withdrawal seizures may occur in epileptic patients withdrawn from phenytoin treatment. In patients with status epilepticus, alcohol and noncompliance with anticonvulsant regimens may be synergistic.[20,38]

Alcoholic patients with preexisting seizure disorders pose a dilemma when they are supposed to be taking antiepileptic drugs (AEDs) but their blood levels suggest

noncompliance. This situation is especially problematic when their epileptic attacks are uncommon and seem to occur exclusively in the context of alcohol withdrawal. Some of these patients may have AWS and may have been misdiagnosed. Others may have a seizure disorder that seems to be confined to the setting of alcohol withdrawal. Such patients have demonstrated that they cannot maintain compliance with their treatment.

A patient taking an AED for an antecedent seizure disorder, which presents with a seizure while intoxicated, falls into a different category. Such an episode could be an isolated event in a usually compliant patient without a history of chronic alcohol abuse. In this patient, a seizure in the setting of a subtherapeutic AED level may represent the consequences of noncompliance with the AED versus AWS.[39]

FUTURE DIRECTIONS

In this age of computerized medicine, it will soon be possible to have a complete medical history on even the most obtunded ED patient. This advancement will go a long way in helping the emergency physician determine if the alcohol-related seizure is an isolated event or a recurrent problem. It will help determine if the patient an epileptic whose presentation is complicated by alcohol or if alcohol is the sole problem. This kind of comprehensive history should help reduce the cost of emergency medicine care by reducing the number of imaging studies and other tests that are currently deemed necessary.

Moreover, as researchers continue to evaluate genetic markers, which may demonstrate a predisposition to ARS, and easily measured biomarkers, such as serum prolactin and homocysteine levels, it may one day be possible to accurately predict which patients are at risk for withdrawal seizures. Therefore, it may one day be possible to accurately separate patients in need of prophylaxis from those in whom seizure treatment is unnecessary.

SUMMARY

Alcohol kills the person who consumes it and it kills unintended victims by the acts of inebriated persons. Whatever medical, traumatic, psychological, or social problem brings an alcoholic patient to the ED, the underlying problem is alcoholism and the ultimate goal is abstinence. Most municipalities have either an Alcoholics Anonymous chapter or a treatment center for anyone who desires help with alcohol. In small communities, clergy or social workers can usually arrange rehabilitation programs. This disease will surely progress if alcoholism is not recognized and the patient is not given the opportunity to participate in a rehabilitation program. It is up to the emergency physician to intervene on behalf of the patient and the public.[40,41]

KEY CONCEPTS

- Not all ARS are AWS
- Search for reversible causes (eg, hypoglycemia, intracranial hemorrhage, CNS infection) in alcoholic patients with seizures
- Obtain a CT scan of the head in alcoholic patients with a seizure of undetermined cause
- Lorazepam is the drug of choice in alcoholic patients who have had a seizure or who are demonstrating signs of withdrawal
- Alcoholic patients who have had a seizure should generally be observed for 4 to 6 hours and, if discharged, referred to a detoxificaton/rehabilitation program, if possible

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